

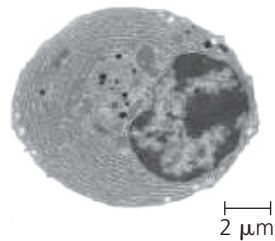
For B cells, the effector forms are *plasma cells* (Figure 43.15), which secrete antibodies. For T cells, the effector forms are helper T cells and cytotoxic T cells, whose roles we'll explore in Concept 43.3. The remaining cells in the clone become **memory cells**, long-lived cells that can give rise to effector cells if the same antigen is encountered later in the animal's life.

Figure 43.16 summarizes the proliferation of lymphocyte into a clone of cells occurs in response to binding to an antigen, using B cells as an example. The process is called **clonal selection** because an encounter with an antigen *selects* which lymphocyte will divide to produce a *clonal* population of thousands of cells specific for a particular epitope.

Immunological Memory

Immunological memory is responsible for the long-term protection that a prior infection provides against many

▼ Figure 43.15 A plasma cell.

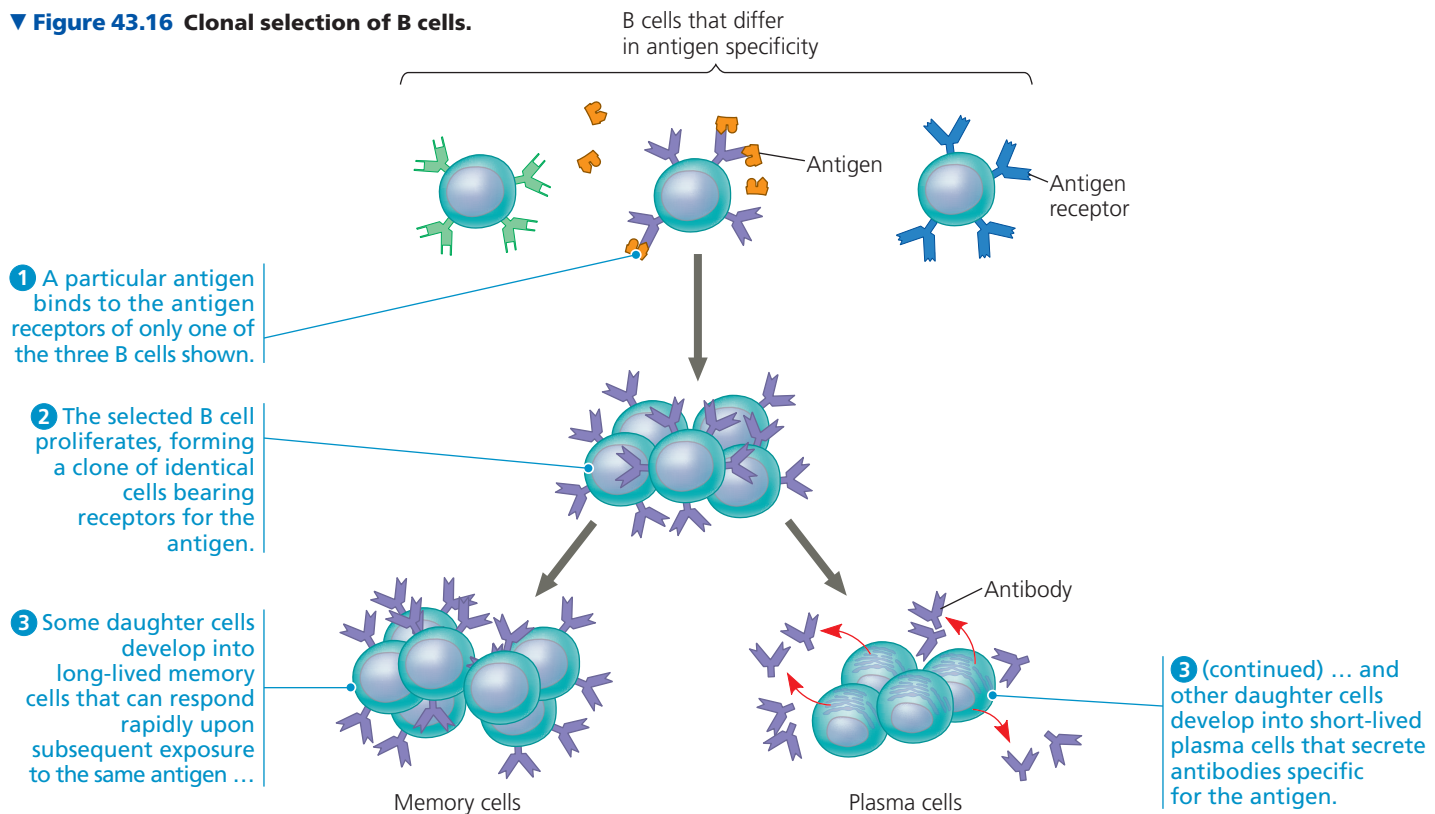


diseases, such as chicken pox. This type of protection was noted almost 2,400 years ago by the Greek historian Thucydides. He observed that individuals who had recovered from the plague could safely care for those who were sick or dying, “for the same man was never attacked twice—never at least fatally.”

Prior exposure to an antigen alters the speed, strength, and duration of the immune response. The effector cells formed by clones of lymphocytes after an initial exposure to an antigen produce a **primary immune response**. The primary response peaks about 10–17 days after the initial exposure. If the same antigen is encountered again later, there is a **secondary immune response**, a response that is faster (typically peaking only 2–7 days after exposure), of greater magnitude, and more prolonged. These differences between primary and secondary immune responses are readily apparent in a graph of the concentrations of specific antibodies in blood over time (Figure 43.17).

The secondary immune response relies on the reservoir of T and B memory cells generated upon initial exposure to an antigen. Because these cells are long-lived, they provide the basis for immunological memory, which can span many

▼ Figure 43.16 Clonal selection of B cells.



VISUAL SKILLS For the purpose of illustration, this figure shows only a few of each type of cell or molecule. Based on what you have read in Concept 43.2, provide estimates of the number of different B cells and number of antigen receptors on each B cell.

➡ Mastering Biology Animation: Clonal Selection